

Manan Sethia

Computer Science undergraduate at NTU Singapore, interested in data engineering, systems, cybersecurity, and practical research tooling.

CONTACT

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TECHNICAL SKILLS

LANGUAGES & TOOLCHAINS

Python, C/C++, Java, JavaScript/TypeScript, SQL, C#/NET, Kotlin, Swift; npm/npv, Maven/Gradle, NuGet, SPM

DATA ENGINEERING

ETL, REST APIs, async/batch ingestion, entity resolution, schema normalization, pandas, SQLite, Excel/CSV; SciVal, Scopus, OpenAlex, ROR, Wikidata, ORCID

AI & DEVELOPMENT

NumPy, scikit-learn, OpenCV, PyTorch basics, Flask, Django, Node.js, RAG, MCP, pytest, Playwright

DEVOPS & LINUX

Git, GitHub Actions/CI/CD, Docker, Bash/zsh, SSH, cron, Nginx; Ubuntu, Debian, Fedora, Arch and Kali

APPS, SECURITY & IOT

Android Studio, Xcode, Wireshark, Burp Suite, Nmap; Raspberry Pi, ESP32, Arduino, sensors, robotics

CORE CS

DSA, competitive programming, OOP, DBMS, systems fundamentals

PROFESSIONAL SKILLS

Team leadership, event management, mentorship, public speaking, presentation, academic writing

EDUCATION

Nanyang Technological University

Bachelor of Computing (Hons) in CS
2026 – 2030 | Singapore
Specialisations: Cyber Security, AI
NTU Science and Technology Undergraduate Scholar

Krishna Public School, Kamal Vihar

Grade 1 – 12 | 2014 – 2026
Raipur, Chhattisgarh, India
Stream: PCME + Computer Science

EXPERIENCE

- Nanyang Technological University** *Jul 2026 – Present*
 - Technical Director, TGIFHacks (NTU OSS)** *Sep 2026 – Present*
Direct technical planning for hackathons, hardware and software builds, workshops, and competitions.
 - Technical Subcommittee Member, NTU Sentinels** *Aug 2026 – Present*
Contribute to cybersecurity and capture-the-flag activities.
 - Data Analytics & Engineering Assistant** *Jul 2026 – Present*
Build auditable Python ETL and identity-resolution pipelines for the Office of Data and Analytics under the Work Study Scheme (ad hoc through 31 Mar 2027). Work spans bibliometric APIs, rate-limit handling, schema normalization, deduplication, manual verification, and SQL/CSV/XLSX delivery. Current datasets include the 45,801-record Stanford Top 2% collection and UBC researcher matching.
- Buildabot Labs Private Limited** *May 2026 – Aug 2026*
 - Chief Technology and Innovations Officer (CTIO)**
Led technology strategy and engineering at FSTI's technology subsidiary. Built database and backend systems, the company website, a RAG assistant, and an emulation playground for virtual project testing.
- First Step Innovations (FSTI)** *Apr 2026 – Jul 2026*
 - R&D Intern**
Worked on full-stack projects: e-commerce web app, AI career counselling platform, and drone assembly simulator. Handled technical documentation, QA testing, and daily progress reporting.
- Krishna Public School, Kamal Vihar** *Apr 2016 – Mar 2026*
 - Headboy, Student Council** *Jul 2025 – Feb 2026*
Led student council overseeing 18 clubs, editorial board, and 97 members. Managed XPLOA, KPSMUN, and Annual Function.
 - Problem Setter, I-Code** *Aug 2024 – Oct 2025*
Designed algorithm and logic questions for coding competitions across difficulty levels.
 - Advisor & Coordinator, Xplora** *Nov 2022 – Nov 2025*
Coordinated flagship tech fest with robowars, ideathons, and gaming; managed outreach and finances.
 - Innovator & Mentor, ATL Club** *Jul 2022 – Mar 2026*
Built prototypes with embedded systems; mentored juniors on robotics and IoT projects.
 - Technical Lead, Coder's Club** *Jul 2019 – Mar 2026*

Member → Vice President → President → Mentor

Hosted workshops on CP, ML, and cybersecurity; championed Linux adoption; guided students.
 - Quiz Organizer & Mentor, QPL** *Apr 2016 – Mar 2026*
Represented school in national quizzes; organized intraschool events across diverse domains.

PROJECTS

MRIDA | Multipurpose Real-time Integrated Digital Analyzer | 2023 – 2025

AI-powered Raspberry Pi soil-health analyzer with 8-in-1 sensing, 4G/GPS, emergency alerts, and speech guidance. Winner at Data & AI Talks; selected for Youth Ideathon Top 1,000, INSPIRE MANAK DLEPC, and the Junior SIH national presentation round.

MATRIX | Desktop Voice Assistant | 2021 – 2022

JARVIS-inspired assistant with Python, Tkinter. Speech/text interaction, weather, news, and hand-drawn GUI.

CERTIFICATIONS

- Google Cybersecurity** Professional Certificate (Coursera)
- IBM Ethical Hacking** with Open-Source Tools (Coursera)
- Programming and DSA** Courses by CodeChef for:
 - Python
 - Java OOP
 - C/C++ with STL

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Curriculum Vitae (contd.)

ACHIEVEMENTS

International

Gold Award (ESP) and Bronze Medal (Individual) at IESO 2024 in Beijing; represented India (Aug 2024)

Selected for **BeyondQuantum 2025**, a 12-week international research program on quantum computing (Mar 2025)

Felicitated by HBCSE-TIFR as an International Olympiad medalist; scholarship recipient (Dec 2024)

- **High Distinction** in the International Chemistry Quiz by RACI, Australia (Mar 2024)

Programming, Innovation & Computing

AIR 10 in HPE Codewars 2025 among 6,000+ participants (Apr 2025)

AIR 17 in IIT Madras Shaastra Programming Contest 2026 among 18,000+ participants (Dec 2025)

National Finalist, Shaastra CP Potpourri 2026 (IIT Madras) (Dec 2025)

Silver Medalist, Future Tech Olympiad 2026 by 1M1B and IBM SkillsBuild (Feb 2026)

Winner, Code 2 Champions at ITM University Raipur; only school student (Apr 2025)

Winner, Coding Competition at Nexus 2K24 bySSIPMT; solved finals in 4 minutes (Nov 2024)

Champion, Data and AI Talks 2023 by iGebra (Aug 2023)

- **Top 1,000** nationally, Youth Ideathon 2023 (CBSE + ThinkStartup) (Oct 2023)
- **Qualified for DLEPC**, INSPIRE MANAK Awards 2023 by DST; received project-development support (Feb 2024)
- Shortlisted for **National Rounds**, Junior Smart India Hackathon 2023 (Jan 2024)

National Olympiads

Qualified for National Camp, INESO 2025 (Mar 2025)

Zonal Topper, ZCO 2024; qualified for INOI (Dec 2024)

Stage 2 Qualifier, INAIO 2025 by ACM India (Jan 2025)

Quizzing

Winner, Clash of Campus Quiz at IIIT Naya Raipur Technovate 7.0; only school participant (Mar 2026)

2nd Runner-Up, Wonga Wits Finance Quiz S2 by Yes Securities; only school student (Nov 2025)

1st Runner-Up Nagpur, TCS InQuizitive 2025; only CG team (Aug 2025)

Winner IIM Bhilai, Brahmaprakash Memorial Materials Quiz 2025; qualified Nationals (Aug 2025)

Winner, Bhopal Region, SBI Scholar Quiz 2025 (Jun 2025)

- **Winner**, Earth Day Quiz 2025 by NEHU Shillong (Apr 2025)

1st Runner-Up Nagpur, TCS InQuizitive 2024 (Nov 2024)

- **1st Runner-Up**, Technical Quiz at Nexus 2K24 (Nov 2024)

- **Winner**, GK Quiz at Mindspark 2024 (Oct 2024)

- **Special Mention (UNGA)**, KPS MUN 3.0 (Oct 2024)

1st Central Zone, SEBI Financial Literacy Quiz 2024 (May 2024)

Runner-Up Bhubaneswar, CBSE Heritage India Quiz 2023 (Dec 2023)

- **Winner**, Tech Quiz at Nexus 2K23 (Nov 2023)

National 1st Runner-Up, Ultimate Sports Quiz S2; aired on JioHotstar (Sep 2023)

All India Top 5, Mind Wars 2023 (May 2023)

National 1st Runner-Up, CBSE Heritage India Quiz 2022; New Delhi finals (Jan 2023)

- **Winner**, Paryavaran Paheli ECO Quiz 2022 (Nov 2022)
- **City Champions Raipur**, INTACH Heritage Quiz 2022 (Oct 2022)
- **Winner**, Shine COSMO GK Quiz 2022 (Sep 2022)

State Champions CG, Fit India Quiz Season 1 (Jun 2022)

- **National Qualifier**, WWF Wild Wisdom 2021 (Oct 2021)